Handling Non-L^1 Signals in Electrical Engineering Fourier Transforms

EE Fourier Analysis Notes

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1 Introduction

In electrical engineering, we frequently encounter foundational signals like the unit step $u(t)$, sinusoids $\cos(\omega_0 t)$, and constants that are not absolutely integrable (L^1). Consequently, their classical Fourier integrals fail to converge in the ordinary sense of calculus. To bypass this, engineering textbooks employ specific algebraic and analytical techniques to achieve valid solutions.

2 Practical Textbook Techniques

Rather than diving immediately into advanced measure theory, textbooks rely on three primary operational workarounds to evaluate these non-L^1 integrals:

Technique A: Convergence Factors (The Real-Part Regulator)

When an integral fails to decay at infinity, we introduce a decaying exponential regulator $e^{-\epsilon|t|}$ where $\epsilon > 0$. We compute the standard convergent integral first, and then take the limit as $\epsilon \to 0^+$. For example, the Fourier transform of the unit step $u(t)$ is handled via:

$$\mathcal{F}\{u(t)\} = \lim_{\epsilon \to 0^+} \int_0^\infty e^{-\epsilon t} e^{-j\omega t}\, dt = \lim_{\epsilon \to 0^+} \frac{1}{\epsilon + j\omega} = \frac{1}{j\omega} + \pi\delta(\omega) \tag{1}$$

Technique B: Duality and Inverse Transforms

Evaluating the forward integral of a constant or a pure sinusoid directly is highly problematic. Instead, we propose a solution involving the Dirac delta function $\delta(\omega)$ and verify it using the inverse Fourier transform. By utilizing the sifting property:

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} 2\pi\delta(\omega) e^{j\omega t}\, d\omega = e^{j(0)t} = 1 \tag{2}$$

By duality, this rigorously establishes that $\mathcal{F}\{1\} = 2\pi\delta(\omega)$.

Technique C: Cauchy Principal Value

For signals containing internal singularities that blow up (such as $1/t$), ordinary integration is undefined. We resolve this by integrating symmetrically around the singularity and taking a limit, known as the Cauchy Principal Value (P.V.):

$$\text{P.V.} \int_{-\infty}^{\infty} \frac{e^{-j\omega t}}{t}\, dt = -j\pi\, \text{sgn}(\omega) \tag{3}$$

3 A Useful Integration-by-Parts Formula

Integrals containing both an exponential and a sinusoid occur frequently in Fourier and Laplace transform calculations. Let

$$I = \int e^{ax} \sin x \, dx,$$

where a is a real constant. Integrating by parts with $u = e^{ax}$ and $dv = \sin x \, dx$ gives

$$I = -e^{ax} \cos x + a \int e^{ax} \cos x \, dx. \tag{4}$$

Apply integration by parts again to the remaining integral, now using $u = e^{ax}$ and $dv = \cos x \, dx$:

$$\int e^{ax} \cos x \, dx = e^{ax} \sin x - aI. \tag{5}$$

Substituting (5) into (4) returns to the original integral:

$$I = -e^{ax} \cos x + ae^{ax} \sin x - a^2 I.$$

Therefore,

$$\boxed{\int e^{ax} \sin x \, dx = \frac{e^{ax}}{a^2 + 1}\big(a \sin x - \cos x\big) + C.} \tag{6}$$

This formula also covers $a = 0$, giving $\int \sin x \, dx = -\cos x + C$. As a check, differentiating the expression on the right side of (6) gives $e^{ax} \sin x$.

4 The Unified Viewpoint: Generalized Fourier Transforms

While Techniques A, B, and C can feel like a disjointed collection of distinct mathematical tricks, they are actually localized shortcuts to a broader, rigorous framework known as the **Generalized Fourier Transform** operating within the space of *Tempered Distributions*.

Learning full distribution theory is significantly harder and mathematically more demanding than standard engineering calculus. However, it provides the ultimate justification for these methods: every single one of these textbook shortcuts is completely sound and ultimately leads to the exact same, mathematically rigorous generalized answer.

5 Homework Exercises

Use the Fourier-transform convention

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} \, dt, \qquad x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} \, d\omega.$$

1. Use the inverse Fourier transform and the sifting property to verify

$$\mathcal{F}\{e^{j\omega_0 t}\} = 2\pi\delta(\omega - \omega_0).$$

2. Write $\cos(\omega_0 t)$ in terms of complex exponentials and find its generalized Fourier transform.

3. Write $\sin(\omega_0 t)$ in terms of complex exponentials and find its generalized Fourier transform.

4. For $\epsilon > 0$, evaluate

$$\int_0^{\infty} e^{-\epsilon t} e^{-j\omega t} \, dt.$$

What expression is obtained before taking the limit $\epsilon \to 0^+$?

5. Let $x(t) = 3 + 2\cos(5t)$. Find $X(\omega)$ as a sum of Dirac delta functions.

6. The function $1/t$ is odd. Explain briefly why a symmetric principal-value integral of $1/t$ over $[-A, A]$ is zero, even though the two ordinary improper integrals on either side of $t = 0$ do not converge.

6 Answer Key

1. Taking the inverse transform of the proposed answer gives

$$\frac{1}{2\pi}\int_{-\infty}^{\infty} 2\pi\delta(\omega-\omega_0)e^{j\omega t}\,d\omega = e^{j\omega_0 t}.$$

2. Since $\cos(\omega_0 t) = \frac{1}{2}(e^{j\omega_0 t} + e^{-j\omega_0 t})$,

$$\mathcal{F}\{\cos(\omega_0 t)\} = \pi\big[\delta(\omega-\omega_0) + \delta(\omega+\omega_0)\big].$$

3. Since $\sin(\omega_0 t) = \frac{1}{2j}(e^{j\omega_0 t} - e^{-j\omega_0 t})$,

$$\mathcal{F}\{\sin(\omega_0 t)\} = \frac{\pi}{j}\big[\delta(\omega-\omega_0) - \delta(\omega+\omega_0)\big].$$

4. Direct integration gives

$$\int_0^{\infty} e^{-(\epsilon+j\omega)t}\,dt = \frac{1}{\epsilon+j\omega}.$$

5. By linearity and the transform pairs above,

$$X(\omega) = 6\pi\delta(\omega) + 2\pi\delta(\omega-5) + 2\pi\delta(\omega+5).$$

6. For every $0 < \eta < A$,

$$\int_{-A}^{-\eta}\frac{dt}{t} + \int_{\eta}^{A}\frac{dt}{t} = 0$$

because the two terms cancel by odd symmetry. Their symmetric limit is therefore zero. Each one-sided integral diverges logarithmically, so the corresponding ordinary improper integral does not exist.